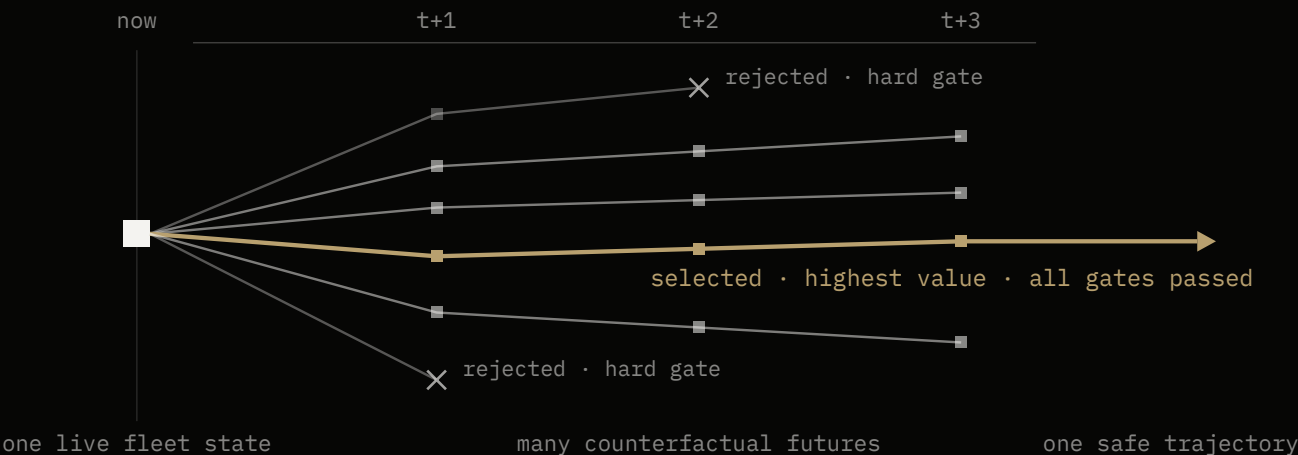


Choose the best fleet trajectory, not just the best next action.

One live fleet state. Many counterfactual futures. One safe trajectory.



Kubernetes, Slurm, serving schedulers, and autoscalers already make and execute control decisions well. Aurelius adds a pre-execution simulation layer above them. Before decisions execute, it forks the live fleet state into competing simulated futures, each applying a different coupled control strategy across batching, routing, placement, capacity, prewarming, admission, migration, precision, and clock.

Each trajectory advances queues, topology pressure, warm state, capacity, SLA risk, and cost through time. Trajectories that violate hard SLA, capacity, power, or policy constraints are rejected. The highest-value feasible trajectory is returned to the existing control plane, which stays authoritative and executes.

8.24×

SLA-safe goodput per dollar vs a production-class composite, uncapped public replay

84.5%

fewer GPU-hours at identical replayed load

1,549,249

replayed requests across three market windows

3 of 3

market windows with better SLA outcomes

Simulated, load-dependent evidence from selected public-trace windows. Not a production result or an estimate for a particular fleet. Full result and scope on page 4.

The optimizer cannot choose a future it never represents.

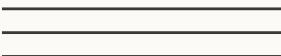
In the ablation behind the architecture, the simulator, forecast, objective, cost model, and safety gates were held constant. Only the set of futures available to the planner changed.



Single-surface search

-4.47%

goodput/\$ vs baseline
SLA gate failed.
173.5% value forfeited vs exhaustive search.



Physics-guided candidates

+100.48%

goodput/\$ vs baseline
SLA gate passed.
30.3% regret vs exhaustive search.



Coupled trajectory search

+161.31%

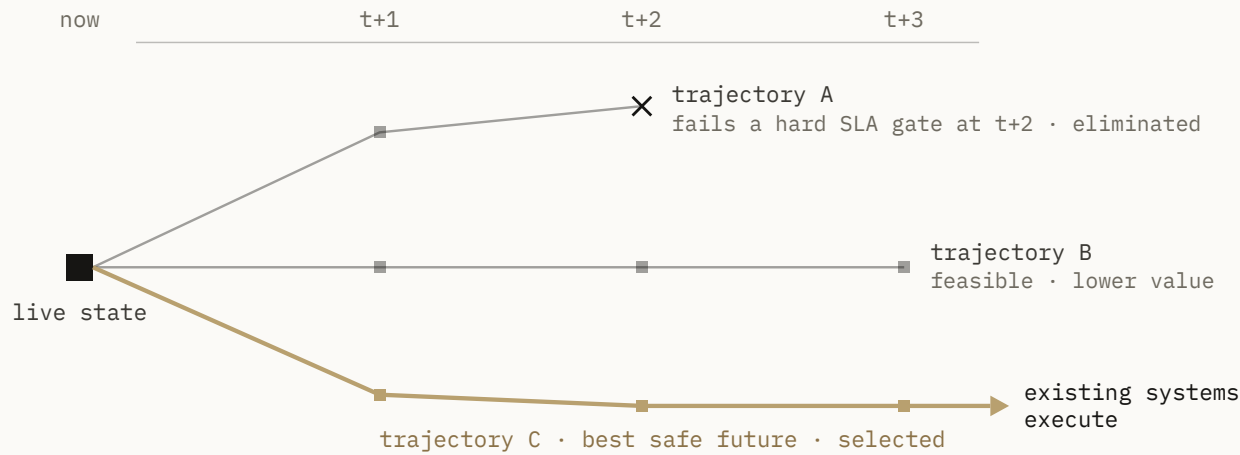
goodput/\$ vs baseline
SLA gate passed.
0% measured search regret vs exhaustive.

The winning strategy combined lower precision, aggressive batching, and a higher clock. No isolated-knob search represented that future: tuned one surface at a time, each move looked worse, so the bundle was never proposed. Once the coupled trajectory entered the candidate set, the unchanged simulator selected it.

The value was not hidden in a better score. It was hidden in a future the previous search never considered.

Benchmark V1, frozen cap-controlled continuity harness, primary market window. Regret is measured against exhaustive enumeration where enumeration is tractable, at roughly 40 trajectory evaluations per decision. Complete methodology in the full technical report.

One fleet. Many futures. Simulated before execution.



Each trajectory advances queues, topology pressure, warm state, capacity, SLA risk, and cost through $t+1$, $t+2$, $t+3$. Every future runs read-only: candidate evaluation never touches the live fleet.

- | | | | |
|---|---|---|---|
| <p>1</p> <p>Mirror the fleet.</p> <p>A persistent world model carries capacity, placement, warm state, queues, migrations in flight, and the operator cost basis, advanced period by period.</p> | <p>2</p> <p>Generate competing coupled trajectories.</p> <p>Physics-guided generation proposes a bounded set of coupled strategies across the control surfaces, never one knob at a time. Known-strong candidates are always retained.</p> | <p>3</p> <p>Run each future forward, read-only.</p> <p>Every candidate trajectory is simulated against forecast conditions using only information available at the decision point. No future truth leaks in.</p> | <p>4</p> <p>Reject unsafe futures. Return the winner.</p> <p>Hard SLA, capacity, power, and policy gates eliminate trajectories outright. The best feasible future is recommended, with a deterministic fallback when confidence is low.</p> |
|---|---|---|---|

Aurelius sits above Kubernetes, Slurm, serving schedulers, routing systems, autoscalers, and fleet-management systems. It coordinates their decisions against a common anticipated future. It does not replace them. It reads scheduler metadata only, never payloads, model outputs, or weights.

A signal too large to ignore and too simulated to trust without operator data.



SLA-safe goodput per dollar vs the constructed production-class composite. Same replayed load per window, 1,549,249 requests total, 84.5% fewer GPU-hours, better SLA in all three windows.

What this establishes

Predictive coupled planning found substantially more efficient safe trajectories than the constructed public-replay composite under the selected uncapped windows.

What this does not establish

Expected performance against a particular fleet or internal scheduler. The result is simulated and load-dependent, and its magnitude is sensitive partly to modeled precision and batching behavior.

Test one bounded historical window.

Input	Anonymized scheduler metadata from one representative workload or cluster window. No customer identity, payloads, model outputs, weights, production actuation, or live integration.
Process	Reconstruct the historical baseline and evaluate Aurelius counterfactual trajectories under identical recorded conditions, with a protocol fixed before any data moves.
Output	Matched SLA, GPU-hour, utilization, and economic comparisons; simulator-fidelity checks; sensitivity analysis; failure cases and negative findings; and a verdict on whether shadow mode is warranted.

The ask

Identify one technically appropriate owner and one bounded historical dataset on which this can be tested.

Beckett Zahedi	Founder · Aurelius	beckett@runaurelius.com	runaurelius.com	Complete methodology, citations, limitations, and version history are in the supporting full technical report.
----------------	--------------------	--	---	--